KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

CHAPTER 2

YEAR 10 MATHEMATICS

Measurement

7 lessons · two pages each · Easy, then Medium

2.1 Converting units
2.2 Perimeter
2.3 Circumference
2.4 Area
2.5 Area of circles
2.6 Surface area
2.7 Volume and capacity

NAME _____ CLASS _____ TEACHER _____

Artwork: feathery gold spirals deep in Elephant Valley, coloured in gold and green, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **two pages**. Read the **learning intention** and tick each **success criterion** when you can do it.

2. Page 1 is **Easy** and page 2 is **Medium**. Each page starts with an example, then questions for you.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Work down the box one round at a time. Answers go on the line; show working in the boxes.

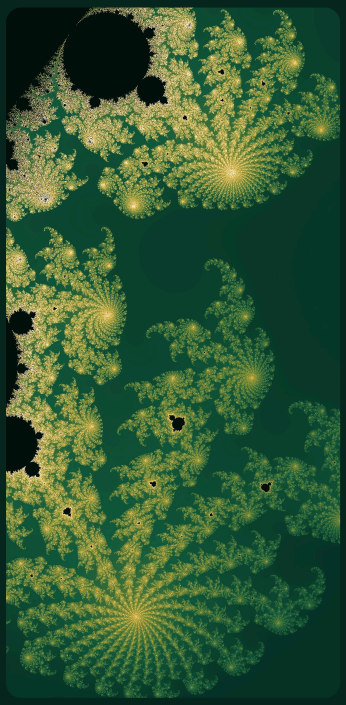
3. Stuck? Read the **Remember** line at the top of the blue box, then look back at the example you copied.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
2.1	Converting units	3		☐
2.2	Perimeter	5		☐
2.3	Circumference	7		☐
2.4	Area	3		☐
2.5	Area of circles	11		☐
2.6	Surface area	13		☐
2.7	Volume and capacity	15		☐

In this chapter you will

- convert between units of length, mass, capacity and time
- find the perimeter of shapes, including composite shapes
- find the circumference and the area of circles
- find the area of squares, rectangles, triangles, parallelograms, trapeziums and rhombuses
- find the surface area of prisms
- find the volume and capacity of prisms and cylinders



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is feathery gold spirals deep in Elephant Valley, coloured in gold and green. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 2.4

Area

Date _____

Learning intention
We are learning to find the area of common shapes, from rectangles to trapeziums.

Success criteria
- ☐ I can find the area of squares, rectangles and triangles
- ☐ I can find the area of parallelograms, trapeziums and rhombuses
- ☐ I can write area in square units (cm², m²)

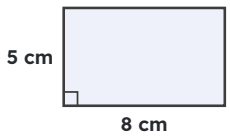
Level 1 · Getting started

Easy · 12 questions

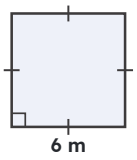
WE DO With your teacher: copy what they write on the board.

Example 1 Find the area of each shape. Area is measured in square units (cm², m²).

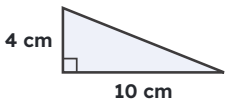
a Rectangle



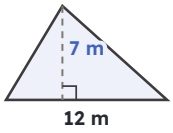
b Square



c Right-angled triangle



d Triangle



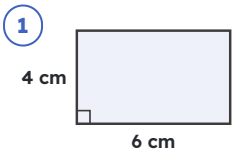
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER Rectangle: A = length × width · Square: A = side × side · Triangle: A = ½ × base × height (the height meets the base at a right angle).

ROUND 1 Questions 1–4: find the area of each rectangle and square.

1



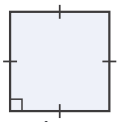
A = _____

2



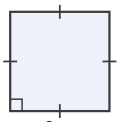
A = _____

3



A = _____

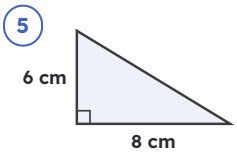
4



A = _____

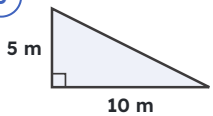
ROUND 2 Questions 5–8: find the area of each right-angled triangle.

5



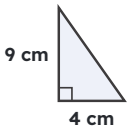
A = _____

6



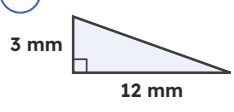
A = _____

7



A = _____

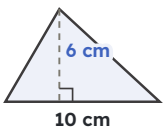
8



A = _____

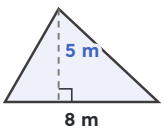
ROUND 3 Questions 9–12: find the area of each triangle. Use the dashed height.

9



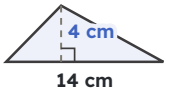
A = _____

10



A = _____

11



A = _____

12



A = _____

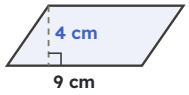
Level 2 · Building up

Medium · 14 questions

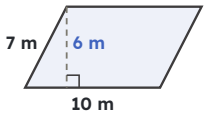
WE DO With your teacher: copy what they write on the board.

Example 2 Use the perpendicular height (the dashed line), not the slanted side.

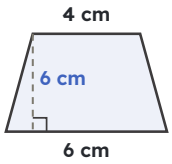
a Parallelogram



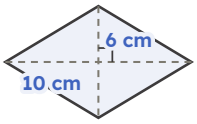
b Parallelogram



c Trapezium



d Rhombus

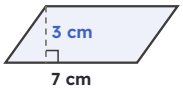


YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 14 QUESTIONS

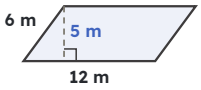
REMEMBER Parallelogram: A = base × height · Trapezium: A = ½ × (a + b) × h · Rhombus: A = ½ × x × y (x and y are the diagonals).

ROUND 1 Questions 1–4: find the area of each parallelogram.

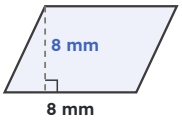
1



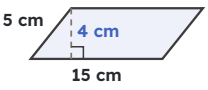
2



3



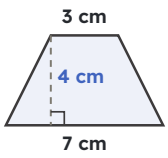
4



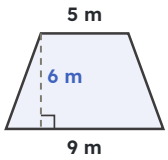
A = _____ A = _____ A = _____ A = _____

ROUND 2 Questions 5–8: find the area of each trapezium.

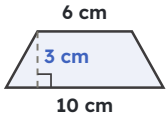
5



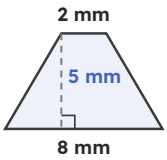
6



7



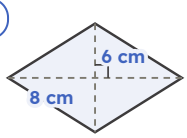
8



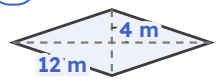
A = _____ A = _____ A = _____ A = _____

ROUND 3 Questions 9–12: find the area of each rhombus.

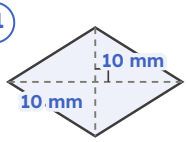
9



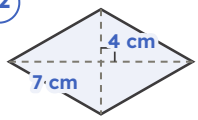
10



11



12



A = _____ A = _____ A = _____ A = _____

ROUND 4 Questions 13–14: solve each problem. Show your working.

13 A rectangular lawn is 12 m long and 8 m wide. Turf costs $15 for each m². How much does it cost to turf the lawn?

14 A triangular sail has a base of 3 m and a height of 5 m. Find the area of the sail.